



EXECUTIVE ANALYTICS REPORT · 2025–2026

UrbanCab Dynamics: Mobility Fare Intelligence & Fleet Analytics

An End-to-End Analytics Pipeline from Raw Ride Logs to Executive Decision Systems

Arunima Mahakul and Ayush Roy

· Python · SQLite · Power BI

Analysis of 1,000 completed trips representing **108,047 ride requests**, **₹370.24K total revenue**, and **17,911 km** traveled across 6 major metropolitan hubs.



The Problem: Demand-Supply Imbalance at Scale

Dynamic ride demand and fleet supply imbalances create extreme surges, transit bottlenecks, and unmet rider demand across urban corridors.

1

Standardize Transaction Logs

Clean and validate messy urban transit records for reliable downstream analysis.

2

Architect a Relational Database

Build a local SQLite database to query operational KPIs and driver rankings.

3

Deliver an Executive Dashboard

Provide interactive Power BI tools for city managers and dispatch teams.

Data Engineering Pipeline: Python QA

```
capstone.py Data Engineering Pipeline - Python Cleaning & QA

1 import pandas as pd
2 import numpy as np
3
4 df = pd.read_csv('UrbanRides.csv')
5 print(df.shape)
6 print(df.head())
7 print(df.tail())
8 print(df.info())
9 print(df.isnull().sum())
10 df['ride_name'] = df['ride_name'].fillna('Unknown')
11 print('Total ride sum:', df['distance_km'].sum())
12 df.columns = [
13     df.columns
14     .str.strip()
15     .str.lower()
16     .str.replace('-', '_')
17 ]
18
19 print(df.columns)
20 test_columns = [
21     'driver_name',
22     'city',
23     'day_of_week',
24     'traffic_level',
25     'type_of_vehicle'
26 ]
27
28 for col in test_columns:
29     df[col] = df[col].astype(str).str.strip()
30
31 print(df['city'].head())
32 print(df['day_of_week'].head())
33 print(df['traffic_level'].head())
34 print(df['type_of_vehicle'].head())
35 df['ride_date'] = pd.to_datetime(
36     df['ride_date'],
37     format='%d-%m-%Y',
38     errors='coerce'
39 )
40 print(df['ride_date'].head())
41 numeric_columns = [
42     'hour_of_day',
43     'distance_km',
44     'no_of_active_drivers',
45     'ride_requests',
46     'ride_distance',
47     'ride_time',
48     'ride_fare',
49     'surge_multiplier',
50     'final_fare'
51 ]
52 print(df[numeric_columns].describe())
53 for col in numeric_columns:
54     negative_count = (df[col] < 0).sum()
55     print(f'{col}: {negative_count} negative values')
56
57 invalid_hours = df[
58     (df['hour_of_day'] < 0) |
59     (df['hour_of_day'] > 23)
60 ]
61 print('Invalid hours:', len(invalid_hours))
62 invalid_distance = df[df['distance_km'] <= 0]
63 print('Invalid distance requests:', len(invalid_distance))
64 invalid_drivers = df[df['no_of_active_drivers'] <= 0]
65 print('Invalid driver requests:', len(invalid_drivers))
66 invalid_requests = df[df['ride_requests'] <= 0]
67
68 print('Invalid ride request records:', len(invalid_requests))
69 df['calculated_ratio'] = 1
70 df['ride_requests'] /=
71 df['no_of_active_drivers']
72
73 df['ratio_difference'] = (
74     df['demand_supply_ratio'] -
75     df['calculated_ratio']
76 ).abs()
77 print(df['ratio_difference'].describe())
78 print(
79     'Ratio mismatches',
80     (df['ratio_difference'] > 0.01).sum()
81 )
82 df['calculated_final_fare'] = (
83     df['base_fare'] *
84     df['surge_multiplier']
85 )
86 df['fare_difference'] = (
87     df['final_fare'] -
88     df['calculated_final_fare']
89 ).abs()
90 print(df['fare_difference'].describe())
91 df['base_fare'] = df['base_fare'].round(2)
92 df['final_fare'] = df['final_fare'].round(2)
93 df['demand_supply_ratio'] = df['demand_supply_ratio'].round(2)
94 df['surge_multiplier'] = df['surge_multiplier'].round(2)
95 print(df.columns)
96 df = df.drop(
97     columns=[
98         'calculated_ratio',
99         'ratio_difference',
100         'calculated_final_fare',
101         'fare_difference'
102     ]
103 )
104 print(df.info())
105 print('===== DATA QUALITY REPORT =====')
106 print('Rows:', len(df))
107 print('Columns:', len(df.columns))
108 print('Duplicate rows:', df.duplicated().sum())
109 print('Missing values:', df.isnull().sum().sum())
110 print('Column data types:')
111 print(df.dtypes)
112
113 print('\nDate range:')
114 print(df['ride_date'].min(), 'to', df['ride_date'].max())
115 df.to_csv('UrbanRides_Cleaned.csv', index=False)
116
117 print('Clean CSV created successfully!')
118 print(df.columns)
119 print('Unique vehicle types:')
120 print(df['type_of_vehicle'].unique())
```

Pre-Processing & Validation

- **Standardized Schema:** Enforced snake_case headers, removed whitespace, and parsed dates to strict DD-MM-YYYY format.
- **Zero-Tolerance Auditing:** Filtered negative fares, checked hour bounds (0–23), and purged zero-distance trips.
- **Mathematical Integrity:** Cross-checked ratios (requests/drivers) and verified final fare logic against surge multipliers.

130

SCRIPT LINES

0

RESIDUAL NULLS

100%

AUDITED CONSISTENCY

Dataset Schema & Feature Structure

AutoSave		urban_rides_cleaned.csv - Excel																		1 / 1	
FileHomeInsertPage LayoutFormulasDataReviewViewReferencesHelp																					
Home		Font	Cells	Style	Formatting	Alignment	Number	Conditional Formatting	Table	Tools											
A1		=A1																			
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P				
1		user_id	driver_name	city	ride_date	hour_of_day	day_of_week	distance_km	traffic_level	type_of_vehicle	no_of_active_drivers	ride_requests	demand_supply_ratio	trip_duration	base_fare	surge_multiplier	final_fare				
2		USDR-21819-GH7	Pedro Sanchez	Pune	2025-11-05	8	Friday	28.21	Low	Motorbikes	35	67	1.86	90	254.52	1.3	400.80				
3		USDR-73403-08w	Maria Rodriguez	Mumbai	2025-10-15	2	Friday	21.08	High	Auto	9	17	1.86	90	282.66	1.3	367.86				
4		USDR-40707-yj5	Sam Cooper	Bengaluru	2025-09-18	8	Sunday	25.54	High	Motorbikes	64	108	1.59	75	344.88	1.3	447.3				
5		USDR-41216-vD0x	Amber Patel	Mumbai	2025-10-15	8	Monday	20.8	High	Auto	25	108	7.52	75	201.6	1.6	322.56				
6		USDR-73403-080x	Kimberly Sanchez	Chennai	2025-09-06	18	Friday	4.48	High	Scor	53	24	9.64	110	120.76	1.6	122.76				
7		USDR-50305-7pp	Thomas Bradley	Delhi	2025-05-16	1	Sunday	29.81	jam	Motorbikes	76	41	9.56	87	348.72	1.6	549.72				
8		USDR-71012-0478	Zachary Taylor	Hyderabad	2025-01-18	9	Monday	29.2	jam	Motorbikes	84	102	1.21	80	390.4	1.1	429.44				
9		USDR-01845-748L	Tricia Valencia	Hyderabad	2025-01-29	8	Friday	24.5	High	Motorbikes	24	25	1.82	87	454.9	1.6	454.0				
10		USDR-60325-580x	Ashley Garrett	Mumbai	2025-07-29	11	Tuesday	6.53	High	Scor	50	63	1.26	22	158.35	1.3	205.87				
11		USDR-43038-78JP	Jack Galloway	Bengaluru	2025-06-11	25	Monday	6.32	Medium	Auto	64	107	1.87	93	126.84	1.3	162.21				
12		USDR-90309-7510	Jordan Baker	Pune	2025-04-11	1	Sunday	8.79	jam	Auto	9	98	1.86	25	155.45	1.6	210.77				
13		USDR-20305-70-8x	George Chapman	Mumbai	2025-09-11	22	Tuesday	11.2	High	Auto	80	120	1.56	30	170.4	1.1	187.44				
14		USDR-25073-762A	Nicholas Lewis	Chennai	2025-10-25	22	Saturday	28.09	jam	Scorion	30	159	4.18	81	231.89	1.1	454.84				
15		USDR-33851-6enn	Scott Brown	Hyderabad	2025-01-26	4	Sunday	18.32	High	Scor	16	20	1.58	84	288.84	1.3	386.76				
16		USDR-61008-70y6	Angela Morton	Mumbai	2025-01-12	15	Friday	21.28	jam	Auto	54	107	1.88	70	285.59	1.3	370.87				
17		USDR-74602-75ean	William Baker	Mumbai	2025-09-06	8	Saturday	24.12	Low	Scorion	10	104	9.68	86	238.56	1.6	445.76				
18		USDR-60201-144M	John Callahan	Bengaluru	2025-04-21	9	Wednesday	15.78	Low	Motorbikes	62	18	9.16	80	229.35	1.6	229.35				
19		USDR-00854-vD7x	Armando Miller	Mumbai	2025-11-12	9	Saturday	18.24	High	Scor	18	178	9.44	54	298.12	1.9	558.42				
20		USDR-99118-vD9g	Taylor Smith	Delhi	2025-07-26	5	Monday	19.54	Medium	Scorion	77	18	9.14	87	282.88	1.6	282.88				
21		USDR-27649-2914x	Lincoln	Delhi	2025-04-18	11	Monday	26.68	Low	Auto	44	71	1.61	67	265.56	1.1	434.12				
22		USDR-44025-vB0Q	Jason Tran	Delhi	2025-02-18	2	Wednesday	24.25	Low	Auto	73	42	9.58	94	461.9	1.6	441.0				
23		USDR-34078-789x	David Smith	Pune	2025-09-27	9	Saturday	16.01	Medium	Motorbikes	83	118	1.44	44	160.12	1.3	208.16				
24		USDR-07400-w48u	Louise Hernandez	Bengaluru	2025-04-16	22	Thursday	11.8	jam	Motorbikes	90	178	1.56	30	170.4	1.2	215.84				
25		USDR-25025-v4DQ	Timothy Hancock	Pune	2025-04-29	2	Sunday	12.49	High	Auto	80	157	1.85	24	164.88	1.1	181.37				
26		USDR-23754-7eeb	Shane Gormon	Pune	2025-04-15	1	Saturday	8.78	High	Auto	9	94	19.44	23	125.35	1.6	210.56				
27		USDR-60344-7pp	Justin Fowler	Delhi	2025-09-26	17	Wednesday	5.5	High	Scor	76	107	2.81	25	149.0	1.6	277.4				
28		USDR-00508-745w	Mrs. Stephanie Lee	Chennai	2025-04-19	5	Friday	23.47	Low	Scorion	30	118	2.56	85	338.87	1.8	595.86				
29		USDR-49465-v6Q	Ayan Gupta	Chennai	2025-10-02	3	Friday	2.06	jam	Auto	86	96	9.88	12	16.72	1.6	51.72				
30		USDR-44069-74P4D	Jennifer Davis	Chennai	2025-01-25	8	Friday	19.22	jam	Auto	22	118	5.26	52	245.76	1.6	393.22				
31		USDR-51345-28q3x	Jose Schaller	Pune	2025-09-05	14	Friday	28.47	Low	Scor	75	35	9.47	89	421.84	1.6	421.84				
32		USDR-67032-75w8	Laura Higgins	Hyderabad	2025-09-27	2	Friday	33.5	jam	Scor	35	52	1.49	95	405.0	1.2	488.0				
33		USDR-62170-22-QH	Melissa Marshall	Mumbai	2025-09-12	1	Saturday	6.6	jam	Motorbikes	6	108	21.8	24	110.2	1.6	110.72				
34		USDR-72774-vp01	Jessica Meadows	Pune	2025-05-24	22	Friday	24.84	Medium	Scorion	76	179	2.36	111	458.88	1.6	748.88				
35		USDR-58122-vy8Q	Dustin Mahara	Hyderabad	2025-05-02	1	Tuesday	26.69	jam	Scorion	74	25	9.54	85	298.35	1.6	298.35				
36		USDR-68088-05w	Antonio Yudi	Chennai	2025-05-19	15	Monday	18.1	jam	Motorbikes	77	58	9.89	47	257.2	1.6	257.2				
37		USDR-73467-04yD	Rebecca Hunt	Pune	2025-04-22	5	Monday	2.23	Medium	Scor	12	102	1.46	150	118.86	1.6	181.84				

UndoCtrl+Z | RedoCtrl+Y | 100% — E +

1,000 Validated Records

→ **Analysis-Ready Core:** 1,000 completed urban trips across 16 fully normalized dimensional and metric attributes.

→ **Feature Depth:** Tracks temporal stamps, traffic density, vehicle class, driver supply, and dynamic surge multipliers.

→ **Downstream Ready:** Verified baseline enabling reproducible reporting across SQLite and Power BI dashboards.

1,000

ROW RECORDS

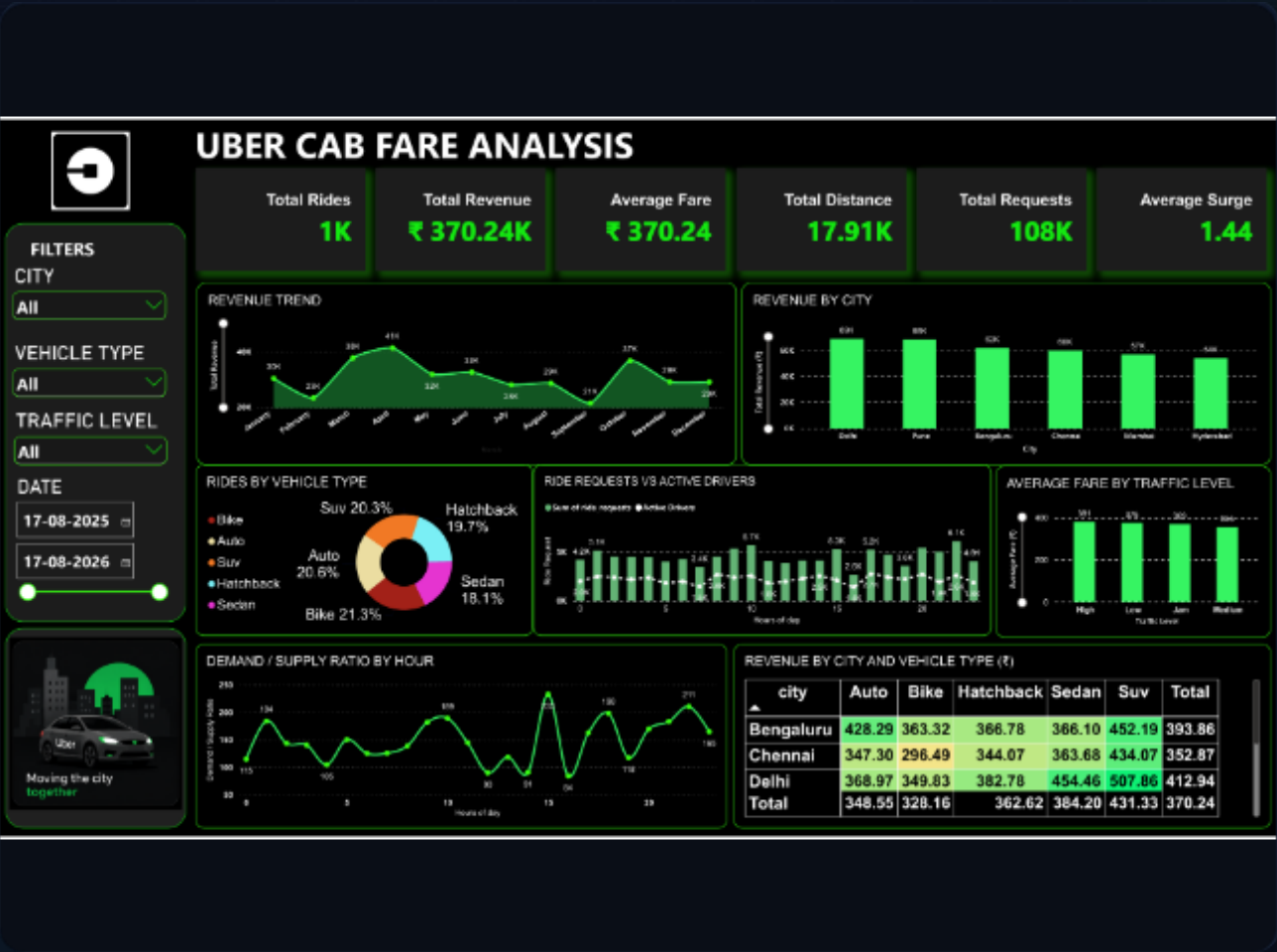
16

FEATURES (A-P)

6

METRO MARKETS

Executive Analytics Control Center



Executive KPI Snapshot

Six primary metrics define operational performance across the full 1,000-ride sample.

1,000

Completed Rides

Total trips in the analyzed sample

₹370K

Gross Revenue

Total revenue realized across all cities

₹ 370

Avg Trip Fare

Mean fare per completed ride

17.9K

Km Covered

Total distance traveled fleet-wide

108K

Market Requests

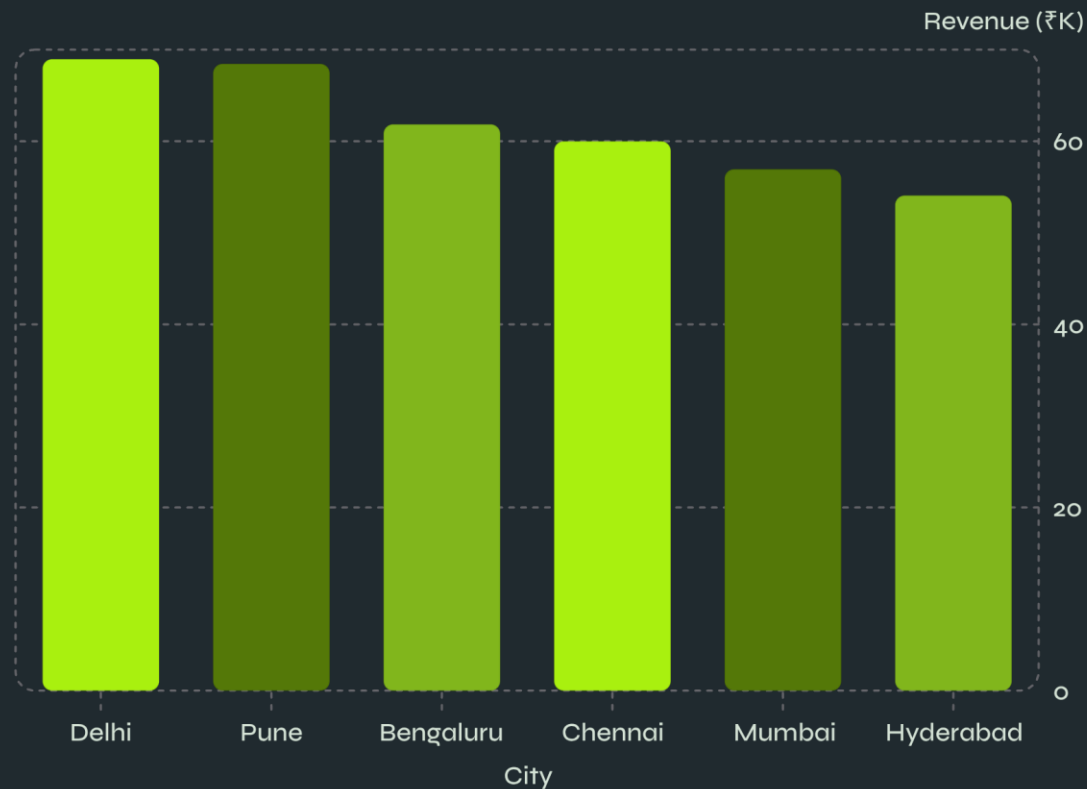
Total ride requests across all hubs

1.44x

Avg Surge

Mean surge multiplier applied

Geographic Market Breakdown: 6-City Revenue Comparison



Strategic Insights

- **Delhi** leads revenue at ₹68.96K with the highest avg fare (₹412.94) across 167 rides
- **Pune** follows at ₹68.45K with the highest ride frequency — 194 completed trips
- **Bengaluru** (₹61.84K) and **Chennai** (₹59.99K) form a strong mid-tier cluster
- **Mumbai** and **Hyderabad** represent the lower-revenue but high-volume tier



Delhi commands the highest ticket size per trip; Pune leads in completed ride frequency.

Advanced SQL Relational Analytics

Enterprise Query Engine

→ **DENSE_RANK() Driver Tiering:** Segments top 3 revenue-generating drivers per city to structure incentive bonuses.

→ **LAG() MoM Revenue Growth:** Calculates consecutive month-over-month revenue deltas across 2025–2026.

→ **NTILE(4) Statistical Quartiles:** Groups ride transactions into 4 fare tiers to isolate top 5% revenue outliers.

34

SQL QUERIES

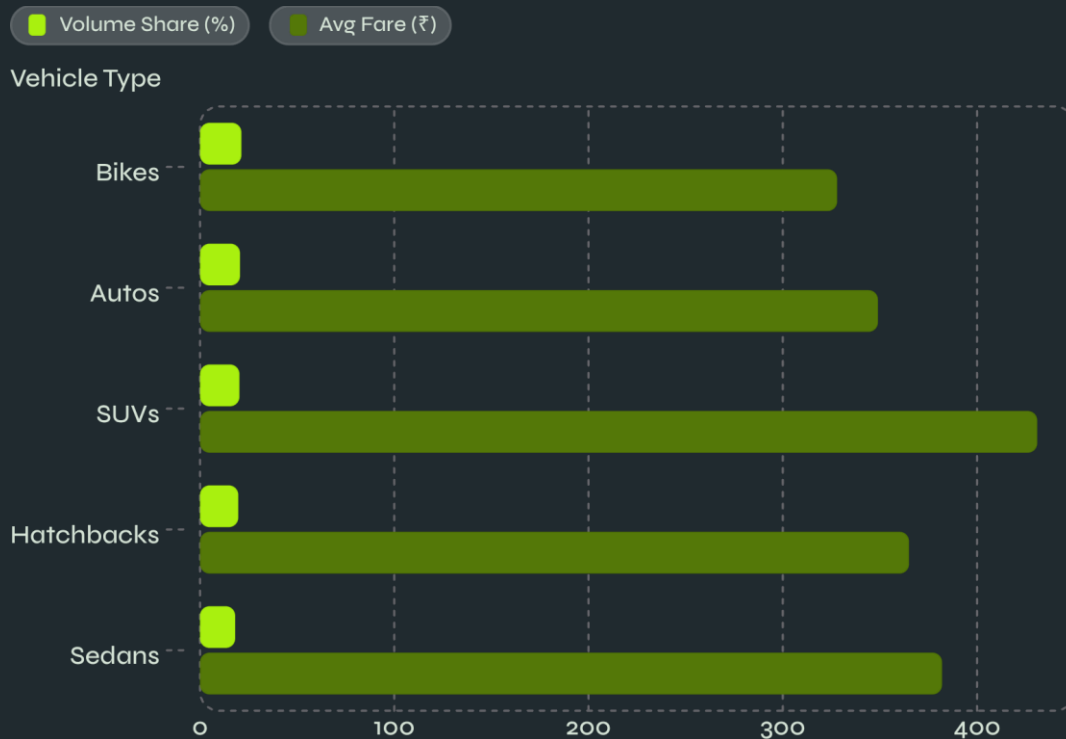
₹112.7K

SURGE REVENUE

30.5%

SURGE MARGIN

Fleet Distribution & Vehicle Economics



Volume vs. Value

- **SUVs** generate the highest individual fares — averaging ₹431.33 across all cities
- **Bikes** and **Autos** provide baseline volume and urban affordability at ₹328.16 and ₹348.55
- Fleet share is remarkably balanced: all five vehicle types sit within a 3-point band (18.1%–21.3%)



Premium Benchmark: Delhi SUVs record the highest cross-tab fare at ₹507.86 — a clear premium corridor signal.

Demand vs. Supply Dynamics & Surge Analytics

Peak Deficit Windows

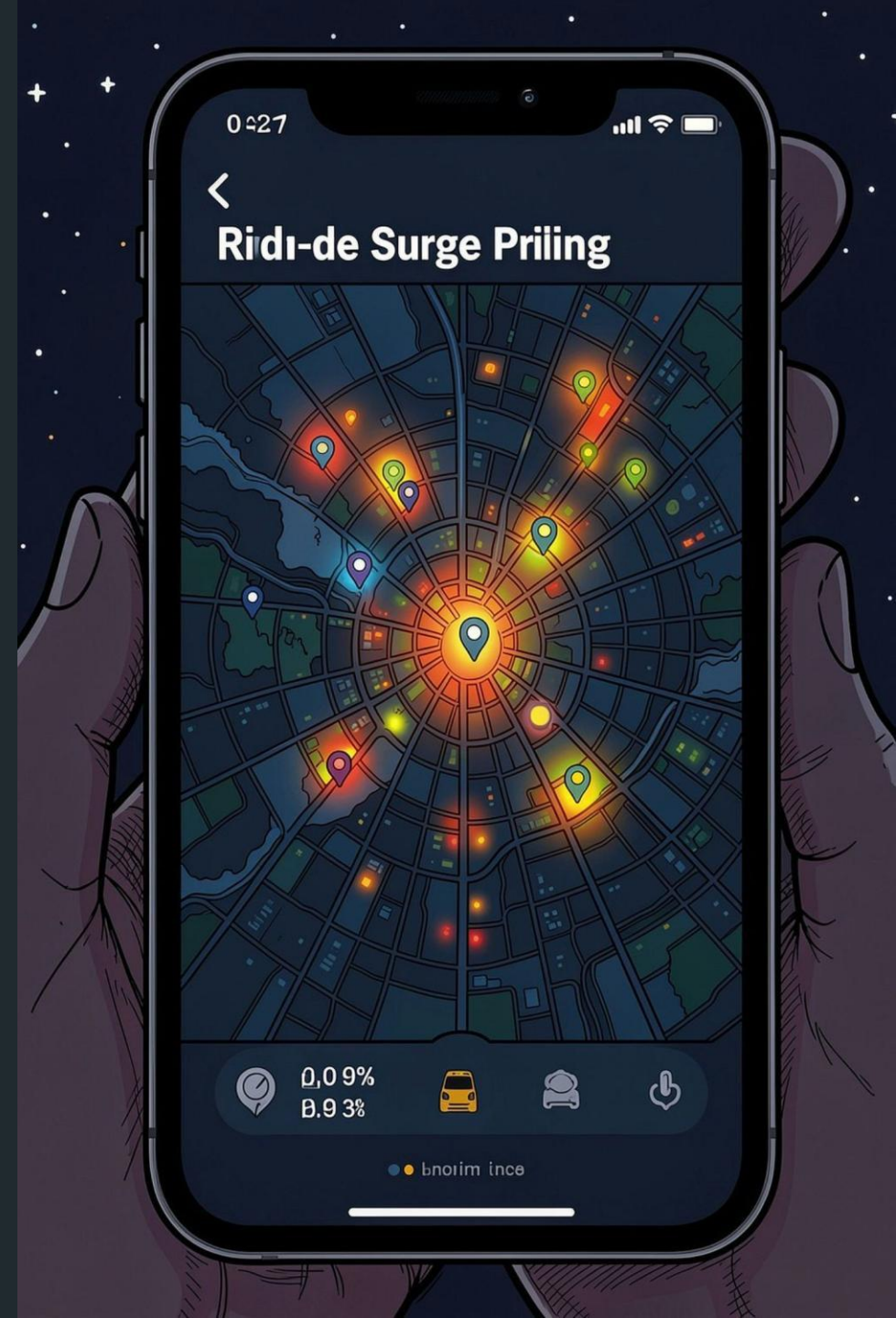
Demand-to-supply spikes between
14:00–16:00 and **21:00–23:00**, reaching ratios of **1.85–2.11x**

₹112.76K Surge Revenue

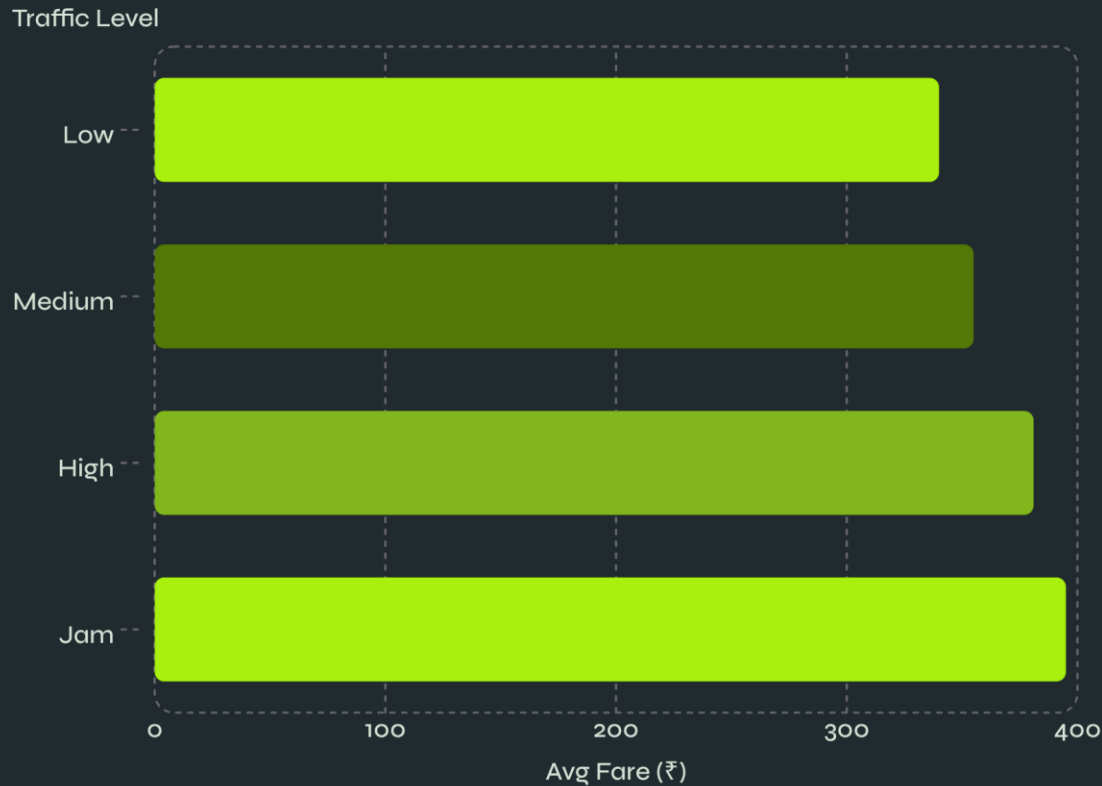
Pure surge pricing contributes
~**30.46%** of total realized
revenue — a major margin driver

Driver Shortage Driver

Surge is driven by **localized driver shortages**, not baseline transit distance — a fleet repositioning opportunity



Traffic Congestion & Operational Velocity



Key Findings

- Traffic levels (Jam → Low) directly influence transit time without suppressing base demand
- **Jam traffic** trips experience speeds <10 km/h and durations exceeding 60 minutes for sub-10 km trips
- Fares average **₹381 in High traffic** vs. **₹355 in Medium** — a 7.3% dynamic pricing response to road delays

📌 Congestion pricing is partially compensating for time- loss, but Jam-tier trips remain operationally inefficient.

Strategic Recommendations & Action Plan



Dynamic Fleet Relocation

Deploy driver repositioning incentives toward peak evening hours (21:00–23:00) where unmet requests peak and supply ratios exceed 2.0x.



Tiered Vehicle Expansion

Expand SUV availability in high-fare corridors — particularly Delhi and Bengaluru — to capture demonstrated premium fare willingness.



Surge Capping & Reliability

Rebalance surge ceilings during peak windows to protect rider retention while preserving driver margin incentives and platform trust.

